

Attack Surface Enumeration Report

1. Indian Railway Catering and Tourism Corporation (IRCTC)

Domains and Subdomains

Primary domain: **irctc.co.in**. Related domains: **irctc.com** and **irctctourism.com**.

- Customer / content: www.irctc.co.in, nget.irctc.co.in, contents.irctc.co.in, care.irctc.co.in, askdishha.irctc.co.in
- Booking / tourism: air.irctc.co.in, bus.irctc.co.in, ecatering.irctc.co.in, tourism.irctc.co.in, hotels.irctc.co.in, rr.irctc.co.in, ftr.irctc.co.in
- Business / operations: operations.irctc.co.in, corporate.irctc.co.in, agent.irctc.co.in, dashboard.irctc.co.in
- Payment-related: pay.irctc.co.in, payment.irctc.co.in, pg.irctc.co.in
- CT also showed environment-style names such as beta.irctc.co.in, uat.irctc.co.in, stageicp.irctc.co.in, stageics.irctc.co.in, stagenget.irctc.co.in, stagews.irctc.co.in, and pmftest.irctc.co.in

CT names are just discovery leads. They do not prove that every host is currently reachable.

IP Addresses / Hosting

Hostname	Observed address / hosting
irctc.co.in	103.252.142.27
www.irctc.co.in	Akamai (edgekey.net / akamaiedge.net); 23.211.142.144, 23.211.142.149 plus IPv6 edge addresses
contents.irctc.co.in	Akamai; 23.211.142.133, 23.211.142.134 plus IPv6 edge addresses
operations.irctc.co.in	103.252.142.41
payment.irctc.co.in	103.252.142.24
agent.irctc.co.in	103.252.142.23
air.irctc.co.in	103.116.163.18
bus.irctc.co.in	103.116.163.54
tourism.irctc.co.in	103.116.163.55
corporate.irctc.co.in	103.116.163.17
care.irctc.co.in	103.116.163.64
ecatering.irctc.co.in	AWS Application Load Balancer in ap-south-1; 13.203.55.152, 15.252.53.247
equery.irctc.co.in	AWS Application Load Balancer in ap-south-1
askdishha.irctc.co.in	Cloudflare Pages
heliyatra.irctc.co.in , lakshadweep.irctc.co.in	Amazon CloudFront
irctc.com	103.116.163.23
irctctourism.com	103.116.163.35

- Authoritative DNS uses Akamai nameservers. Mail routing: mail.irctc.co.in, mail1.irctc.co.in, and mail.nget.irctc.co.in.
- 103.252.142.0/23 is registered to the Centre for Railway Information Systems (CRIS), AS45596.
- APNIC registers 103.116.160.0/22 to IRCTC; public routing sources associate it with AS38799.
- Registry data also showed DNSSEC and registrar locks for irctc.co.in.

Services Identified

- irctc.co.in returned HTTP 302 to the main next-generation e-ticketing application.
- contents.irctc.co.in returned HTTPS 200 and serves public documentation/content.

- ecatering.irctc.co.in returned HTTPS 200 and exposed an nginx server header.
- Third-party indexing associated the agent interface with Java/JSP, PrimeFaces, and jQuery, without a reliable version.
- Shodan reported TCP 80 and 443 on 103.252.142.27 (last seen 1 May 2026), TCP 443 on 103.252.142.41 (22 August 2026), and TCP 80 and 443 on 103.116.163.18 (May 2026).

Exposed Interfaces

- Authorized-agent login at operations.irctc.co.in/AgentInterface/loginHome.jsf, using user ID, password, CAPTCHA, and acceptance of agent rules.
- Public customer booking, tourism, payment, and support portals on the hostnames listed above.

Interesting Findings

- The fetched agent-login page rendered an internal-looking identifier (nget... node/application text) in its footer.
- CT and current DNS together showed several UAT, stage, and test labels that still resolve, including uat, uat.corporate, uat.care, uat.menurates, uatcatering, uatbus, uathotels, pmftest, stagenget, stageicp, stageics, stagemobile, stagews, and testngetjp. No HTTP request was made to these systems.
- Most of the exposed surface was around booking, agent, payment, travel, and support portals.
- No open directory, public bucket, or non-web administrative protocol was verified.

2. Life Insurance Corporation of India (LIC)

Domains and Subdomains

Primary domain: **licindia.in**.

- Public / customer: www.licindia.in, ebiz.licindia.in, eterm.licindia.in, onlinesales.licindia.in, linkpan.licindia.in, pgslink.licindia.in
- Employee / business: hrms.licindia.in, vendors.licindia.in, webmail.licindia.in, efeap.licindia.in
- Platform / integration: digital.licindia.in, portal.licindia.in, nextgen.licindia.in, static.nextgen.licindia.in
- Collaboration: vc.licindia.in, webrtc.vc.licindia.in
- CT also showed names such as internal.licindia.in, sample.access.licindia.in, and origin19953-new.licindia.in

IP Addresses / Hosting

Hostname	Observed address / hosting
licindia.in, www.licindia.in	64.185.181.238 (shared BitGravity / Tata Communications CDN address)
eterm.licindia.in	125.16.45.17
onlinesales.licindia.in	125.16.45.214
hrms.licindia.in	Amazon CloudFront
ebiz.licindia.in	Cloudflare
digital.licindia.in	MoEngage-related CNAME chain fronted by Cloudflare
portal.licindia.in	MoEngage / Amazon CloudFront CNAME chain
linkpan.licindia.in, pgslink.licindia.in	No A / AAAA / CNAME data during the observation window

- 125.16.45.0/24 is described in APNIC-derived data as assigned to LIC and routed by Bharti Airtel (AS9498).
- Authoritative DNS: ns09.domaincontrol.com, ns10.domaincontrol.com.

Services Identified

- licindia.in returned HTTPS 200 and served the official LIC website.

- hrms.licindia.in returned HTTPS 200. LIC's official procurement notice confirms an HRMS implementation, and the main site exposes an Employee's Corner.
- ebiz.licindia.in returned HTTPS 403 through Cloudflare, so the web edge is reachable but access-controlled.
- term.licindia.in returned HTTP 405 to a HEAD request, which just means that method is not allowed.
- digital.licindia.in returned JSON with HTTP 405 through Cloudflare.
- portal.licindia.in returned an AWS ELB-style HTTP 404 at /. That confirms an active endpoint, not a user-facing application at the root path.
- Shodan reported TCP 80 and 443 on the shared CDN IP 64.185.181.238. Extra ports on that shared node cannot be attributed to LIC without hostname-level evidence.

Exposed Interfaces

- Official public website on licindia.in.
- Employee HRMS / Employee's Corner.
- ebiz edge (reachable, but blocked with 403).
- digital and portal endpoints behind Cloudflare / CloudFront.

Interesting Findings

- digital and portal delegate through marketing/CDN infrastructure rather than LIC-owned address space. No dangling-CNAME or takeover condition was established.
- HRMS and Employee's Corner are legitimate employee services, but they do increase the value of LIC-themed phishing.
- linkpan and pgslink have CT history but no current address response.
- No open directory, public bucket, or exposed non-web service was verified.

3. Bharat Sanchar Nigam Limited (BSNL)

Domains and Subdomains

Primary domain: **bsnl.co.in**. Related official service domain: **bsnl.in**.

- Customer service: selfcare.bsnl.co.in, bookmyfiber.bsnl.co.in, portal.bsnl.in, portal2.bsnl.in, mybillview.bsnl.co.in, mobilebillview.bsnl.co.in
- Product / registration: cymn.bsnl.co.in, sancharaadhaar.bsnl.co.in, ngm.bsnl.co.in
- Enterprise / operations: erp.bsnl.co.in, erpportal.bsnl.co.in, fiori.bsnl.co.in, eportal.erp.bsnl.co.in, mail.bsnl.co.in, drive.bsnl.co.in, collab.bsnl.co.in, ptcc.bsnl.co.in
- Hosting / vendor: btcl.bsnl.co.in, bwsbtcl.bsnl.co.in, cloudx.bsnl.co.in, drivex.bsnl.co.in
- Regional: ap.bsnl.co.in, assam.bsnl.co.in, kerala.bsnl.co.in, kolkata.bsnl.co.in, chennai.bsnl.co.in, kl.bsnl.co.in
- CT also included operational names such as intranet.bsnl.co.in, bbnwintranet.bsnl.co.in, nms.mpsnoc.bsnl.co.in, nms.uksnoc.bsnl.co.in, and several devsrv*.kl.bsnl.co.in labels

IP Addresses / Hosting

Hostname	Observed address / hosting
bsnl.co.in	218.248.240.69
bsnl.in	218.248.240.47 and IPv6 2001:4490:effc:3:4000::47
www.bsnl.co.in	CNAME to bsnl.co.in
selfcare.bsnl.co.in	218.248.243.101
portal.bsnl.in, portal2.bsnl.in	218.248.6.241
bookmyfiber.bsnl.co.in	218.248.240.82
cymn.bsnl.co.in, sancharaadhaar.bsnl.co.in	117.236.172.155
mail.bsnl.co.in	164.100.15.5
erpportal.bsnl.co.in, fiori.bsnl.co.in	61.1.112.226
drive.bsnl.co.in	218.248.233.168
collab.bsnl.co.in	218.248.233.169
btcl.bsnl.co.in and related branded hosting	Third-party provider address space
etender.bsnl.co.in	No A / AAAA / CNAME data during the observation window

- Authoritative DNS: ns11.bsnl.in, ns12.bsnl.in.
- Mail routing uses mx.mgovcloud.in, mx2.mgovcloud.in, and mx3.mgovcloud.in.
- 218.248.240.0/20 and 61.0.0.0/15, plus AS9829, are associated with BSNL's National Internet Backbone.
- The webmail address is in NIC space. Selected branded hosting services use external providers. These are supplier dependencies, not evidence of compromise.
- Current DNS published SPF with a soft-fail policy and DMARC at p=quarantine. Registry data reported an unsigned DNSSEC delegation.

Services Identified

- bsnl.co.in returned HTTPS 200 with an Apache header.
- selfcare.bsnl.co.in returned HTTP 302 to home.xhtml. The indexed login page uses username, password, and CAPTCHA.
- portal.bsnl.in and portal2.bsnl.in returned HTTPS 200 with Apache headers and provide customer/payment services.
- bookmyfiber.bsnl.co.in redirected to the current fibre-booking domain.
- sancharaadhaar.bsnl.co.in redirected to a login application.
- erpportal.bsnl.co.in exposed an SAP NetWeaver employee-service login.
- drive.bsnl.co.in exposed an authenticated drive login; no files were publicly listed.
- collab.bsnl.co.in displayed a stock Apache Ubuntu default page.
- Search indexing exposed Tomcat 8.0.46 documentation under ptcc.bsnl.co.in/docs/. Documentation metadata does not prove the running server version.
- Shodan reported TCP 443 on 218.248.243.101, last seen 3 September 2026, and associated several BSNL customer-care hostnames with that address.

Exposed Interfaces

- Self-care login (username, password, CAPTCHA).
- Customer / payment portals on portal.bsnl.in and portal2.bsnl.in.
- Sancharaadhaar login redirect.
- SAP NetWeaver employee-service login on erpportal.bsnl.co.in.

- Authenticated drive login on drive.bsnl.co.in.

Interesting Findings

- BSNL has a large public estate: self-care, billing, registration, regional sites, employee apps, drive/collaboration, and hosting. Operational VPN / NMS / ITSM names are also visible in current CT/DNS. Visibility is not a vulnerability, but it does make targeting and phishing easier.
- A stock Apache page and indexed Tomcat documentation disclose some deployment metadata.
- A request to cymn.bsnl.co.in produced a redirect containing WAF diagnostic fields, including a threat-category label and requesting-client metadata.
- Indexed self-care pages still contain “old browser / IE8 and above” guidance. That does not prove a vulnerable server.
- CT names containing intranet, nms, vpn, and devsrv should be checked against split-DNS and exposure policy. Current DNS for selected operational names only shows discoverability, not unauthorized access.
- Observed email/DNS posture: SPF soft-fail, DMARC p=quarantine, unsigned DNSSEC. These are not standalone vulnerabilities.
- No open directory, public bucket, or exposed database was verified.

4. Airports Authority of India (AAI)

Domains and Subdomains

Primary domain: **aai.aero**.

- Public / services: www.aai.aero, aim-india.aai.aero, digitalsky.aai.aero, onlinefpl.aai.aero, nocas2.aai.aero, gagan.aai.aero
- Employee / retiree: eoffice.aai.aero, ess.aai.aero, rep.aai.aero, ithelpdesk.aai.aero, mail.aai.aero
- Business / operations: epay.aai.aero, etender.aai.aero, boardmeeting.aai.aero, bems.aai.aero, echart.aai.aero
- CT also showed environment-style names such as adss-test.aai.aero, stagenocas2.aai.aero, and legacy.aai.aero

rep.aai.aero and aim-india.aai.aero were found through official/search indexes even though they were not in the bounded CT output used for this report.

IP Addresses / Hosting

Hostname	Observed address / hosting
aai.aero, www.aai.aero	103.142.16.141
rep.aai.aero	103.142.16.141
eeoffice.aai.aero	103.142.16.160
aim-india.aai.aero	103.142.16.192
ess.aai.aero	103.142.16.189
onlinefpl.aai.aero	59.179.29.105
digitalsky.aai.aero	AWS Elastic Load Balancer in ap-south-1; 13.201.249.183, 13.206.59.222
epay.aai.aero, ithelpdesk.aai.aero	No A / AAAA / CNAME data during the observation window

- Authoritative DNS: Cloudflare nameservers.
- Mail routing: mail.aai.aero, mail1.aai.aero.
- 103.142.16.0/24 is associated with AAI, AS133725.

Services Identified

- aai.aero and www.aai.aero returned HTTPS 200.

- eoffice.aai.aero returned HTTP 302 to an authentication path. The indexed page exposes login ID, password, CAPTCHA, and support details.
- rep.aai.aero returned HTTPS 200 and exposes a retired-employee OTP/login service.
- aim-india.aai.aero returned HTTPS 200 and exposes an Aeronautical Information Management login.
- ess.aai.aero returned HTTP 307 to /irj/portal. Public indexing identifies the target as an SAP NetWeaver Portal login.
- digitalsky.aai.aero returned HTTPS 200 with an nginx header.
- onlinefpl.aai.aero returned HTTP 400 to the minimal request, so there is a responding HTTPS endpoint, but this does not establish application health.
- Shodan did not provide current host records for the two checked AAI-owned addresses. That absence should not be treated as proof that ports are closed.

Exposed Interfaces

- Main public website.
- eOffice login (login ID, password, CAPTCHA).
- Retired-employee OTP/login on rep.aai.aero.
- AIM login on aim-india.aai.aero.
- SAP NetWeaver Portal login on ess.aai.aero.
- DigitalSky HTTPS application.

Interesting Findings

- WebTechSurvey reported Drupal 7 and jQuery 2.2 on aai.aero in April 2026. Drupal 7 community security support ended on 5 January 2025. This is a third-party fingerprint, not a version-confirmed vulnerability.
- There are several public credential-entry points: employee, eOffice, SAP ESS, retiree, AIM, and main-site employee logins.
- CT names such as adss-test, stagenocas2, and legacy should be checked against the actual inventory.
- No open directory, public bucket, or exposed database was verified.

Key Observations

- CT records helped identify several additional hostnames, including some older or environment-related names, even when hosts no longer resolve.
- Hosting is mixed: some services sit on the organizations' own networks, others on CDNs, cloud load balancers, government mail, or specialist vendors.
- Most of the exposed surface was around login, customer, employee and partner portals.
- Passive indexes like Shodan are incomplete and can be misleading on shared CDN addresses. The hostname matters more than the IP alone.
- IRCTC's public agent page showed an internal-looking deployment identifier.
- BSNL's cymn redirect showed client-visible WAF diagnostic parameters.
- LIC's digital and portal names resolve through third-party engagement/CDN infrastructure rather than LIC-owned address space.
- AAI's primary site was independently fingerprinted as an older CMS generation. That still needs internal verification.

Potential Risks

- Orphaned subdomains, certificates, or vendor CNAMEs after a project is shut down.
- Credential phishing and stuffing across many branded login pages.

- Inconsistent patching and logging across owned, cloud, CDN, and vendor-operated services.
- Information disclosure through backend names, diagnostic redirects, server headers, and public procurement documents.
- A missing Shodan/Censys result can look safer than it is. External checks need more than one data source.

Limitations

- DNS answers can change with time and location, especially behind CDNs.
- CT records are historical and include inactive names.
- Search engines and Shodan can be stale, incomplete, or wrong.
- A HEAD response may differ from a normal browser GET.
- This was public recon only, not a penetration test.

Sources

Accessed on 4 September 2026 unless a source-specific date is stated.

Certificate Transparency

crt.sh — IRCTC: <https://crt.sh/?q=%25.ircrc.co.in>
crt.sh — LIC: <https://crt.sh/?q=%25.licindia.in>
crt.sh — BSNL: <https://crt.sh/?q=%25.bsnl.co.in>
crt.sh — AAI: <https://crt.sh/?q=%25.aai.aero>

Official pages

IRCTC: <https://www.ircrc.co.in/nget/train-search>, <https://www.ircrc.com/>, <https://www.ircrcrcrcrc.com/>,
<https://www.operations.ircrc.co.in/AgentInterface/loginHome.jsf>, <https://www.air.ircrc.co.in/>
LIC: <https://licindia.in/>, <https://licindia.in/employee-s-corner>,
https://licindia.in/documents/20121/687365/NOTIFICATION_OF_AWARD.pdf/a5625183-660c-376d-e15c-107f8acfdb30
BSNL: <https://selfcare.bsnl.co.in/tungsten/UI/facelets/login.xhtml>, <https://portal.bsnl.in/myportal/authorize.do>, <https://assam.bsnl.co.in/>
AAI: <https://www.aai.aero/>, <https://eoffice.aai.aero/SSOComponent/gate.php>, <https://www.aai.aero/en/user>, <https://rep.aai.aero/>,
https://aim-india.aai.aero/modal_forms/nojs/login, <https://ess.aai.aero/irj/portal>

Passive indexes and network ownership

Shodan: 103.252.142.27, 103.252.142.41, 103.116.163.18, 64.185.181.238, 218.248.243.101
IPinfo / APNIC: IRCTC 103.252.142.27 and 103.116.163.18 ; LIC 125.16.45.0/24 (AS9498) ; BSNL 218.248.240.69 and AS9829 ; AAI 103.142.16.141
BSNL domain registration: <https://rdap.nixiregistry.in/rdap/domain/bsnl.co.in>

Technology fingerprint

WebTechSurvey (AAI): <https://webtechsurvey.com/website/aai.aero>
Drupal 7 end-of-life notice: <https://www.drupal.org/about/announcements/blog/drupal-7-has-reached-end-of-life-psa-2025-01-06>

Conclusion

Public DNS, certificate logs, official links, passive indexes, and a few simple HTTP checks were enough to map the main service boundaries for these four organizations. The interesting parts were mostly login portals, mixed hosting, and a handful of leftover environment names rather than any obviously open storage or database. The Drupal 7 fingerprint on AAI still needs internal confirmation, and a few public pages leaked extra technical detail that is probably not needed. This report is only a recon inventory; it does not prove that any of the listed systems are vulnerable.